

CHAPTER – 1

INTRODUCTION

1.1 INTRODUCTION

Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterized by cognitive decline and structural brain changes. It is a progressive disease beginning with mild memory loss and possibly leading to loss of the ability to carry on a conversation and respond to the environment. Alzheimer's disease involves parts of the brain that control thought, memory, and language. It can seriously affect a person's ability to carry out daily activities. Alzheimer's disease disrupts the brain's natural asymmetry, often causing faster atrophy in the left hemisphere and leading to widespread loss of grey matter, particularly in memory-critical areas like the hippocampus. This imbalance, coupled with the overall decline in grey matter, contributes to the progressive cognitive decline experienced by Alzheimer's patients.

Therefore, we have used a prevalent computer vision technique for Image Classification – Convolutional Neural Networks to identify the differences in the Images of MRI scans among brains of people with no Alzheimer's and people with 3 different stages of early Alzheimer's Disease (Very Mildly Demented, Mildly Demented and Moderately Demented)

1.2 PROBLEM STATEMENT

Identifying Mild Cognitive Impairment at the early stages is crucial to the detection and diagnosis of Alzheimer's Disease (AD). This project aims to accurately detect mild dementia based on MRI images of the cross section of the brain. The future scope of this project is to choose the best CNN model architecture that produces the most accurate results and build an intuitive

interface that can help with the early detection of cognitive impairments using computer vision and tests.

1.3 PROJECT OVERVIEW

This research aims to develop a deep learning model utilizing the ResNet architecture within Convolutional Neural Networks (CNNs) for accurate classification of Alzheimer's disease (AD) from brain MRI images. The objective is to differentiate between healthy individuals and those affected by AD by leveraging the distinctive structural patterns captured by the uniquely optimized architecture. This study addresses the critical need for early and precise diagnosis of AD, facilitating timely interventions and improving patient care.

1.4 OBJECTIVES

1. To design and implement a CNN architecture, for early Alzheimer's detection and classification from MRI images.
2. To curate and preprocess a comprehensive dataset of MRI scans for training and validation.
3. To evaluate the CNN model's accuracy, sensitivity, and specificity through rigorous testing and validation procedures.
4. To incorporate validated cognitive assessment tools and the most accurate model to analyse user responses and predict an individual's risk of developing AD based on MRIs and the clinical test scores.
5. To ensure data security and compliance with privacy regulations in handling patient information.

Chapter – 2

LITERATURE SURVEY

2.1 INTRODUCTION:

This literature study explores existing researches on Computer Vision models for Image Classification in the domain of healthcare. By reviewing research papers, scholarly articles and books, relevant publications, this survey aims to understand the best practices and performances of different techniques to classify images in the medical domain. The primary focus is on the accuracy of the proposed model as gross mistakes in classification may lead to a huge problem in a domain as critical as healthcare.

2.2 LITERATURE REVIEW

There are many architectures to perform image classification using CNNs. The literature survey was conducted to figure out the architectures that produced the most accurate results.

In [1] It is discussed that Deep learning architectures in healthcare systems include Convolutional Neural Network (CNN), Deep Belief Networks (DBN), Auto-Encoder (AE), and Recurrent Neural Network (RNN). These architectures have already been used in detection and prediction of Cancer, multiple sclerosis, Parkinson disease, etc. Deep Learning architecture is defined as an Artificial Neural Network (ANN) with two or more hidden layers for enhancing prediction accuracy. CNN architecture typically consists of two convolutional layers followed by pooling/subsampling layers, fully connected layers, and a final output layer. Hybrid and ensemble methods based on deep learning have shown to produce better accuracy results in healthcare applications compared to single techniques.

From [2] and [5] many CNN model architectures that may be used for image classification were studied and architectures like LeNet-5, AlexNet, VGG16, VGG19 pretrained models, ResNet50, CNN-RNN, SVM and UNet architectures were considered for the classification task but were finally put on hold after thorough inspection and feasibility study.

From [3], we can see a CNN-based algorithm use MRI coronal slices covering the medical temporal lobe to classify AD patients and CN controls. The convolutional neural network (CNN) model used in the study involves a pretrained network architecture (Inception V4) combined with the subject's age, sex, and slice location. The model processes 1 out of 30 coronal slices independently for each subject, with the results of the 30 slices averaged to produce an Alzheimer's disease probability for that subject.

In [6], they have used three pre-trained networks namely LeNet, AlexNet and ResNet-18 are used in classification.

We have decided to use a custom ResNet model with 64 layers and a dense layer of interconnected neurons as the final layer and a CNN-SVM based classification for this study.

2.3 SUMMARY

The Literature review emphasizes the use of CNNs for most of the classification tasks in healthcare. The architectures that are part of the Residual Network (ResNet) family show a high accuracy rate among all other options. The ResNet50 architecture is computationally much more expensive than ResNet18 but it is worth noting the performance of both the models for the required task. Hence, we have decided to use the architectures mentioned in the last section first to see the level of obtainable accuracy and subsequently modify the model to suit the dataset used.

Chapter – 3

DATASET ANALYSIS

3.1 DATASET ANALYSIS

The dataset that we are using for this classification task is taken from Kaggle. **Augmented Alzheimer MRI Dataset V2** is a dataset consisting of MRI images. The data has four classes of images both in training as well as a testing set: Mildly Demented, Moderate Demented, Non-Demented and Very Mildly Demented. The data contains two folders. One of them is augmented ones and the other is original. Original could be used for validation or test dataset. Data is augmented from an existing dataset.

There is a total of 33984 images of size 200x190px classified into 4 labelled categories in the dataset. This proves to be sufficient enough for the training the selected CNN models.

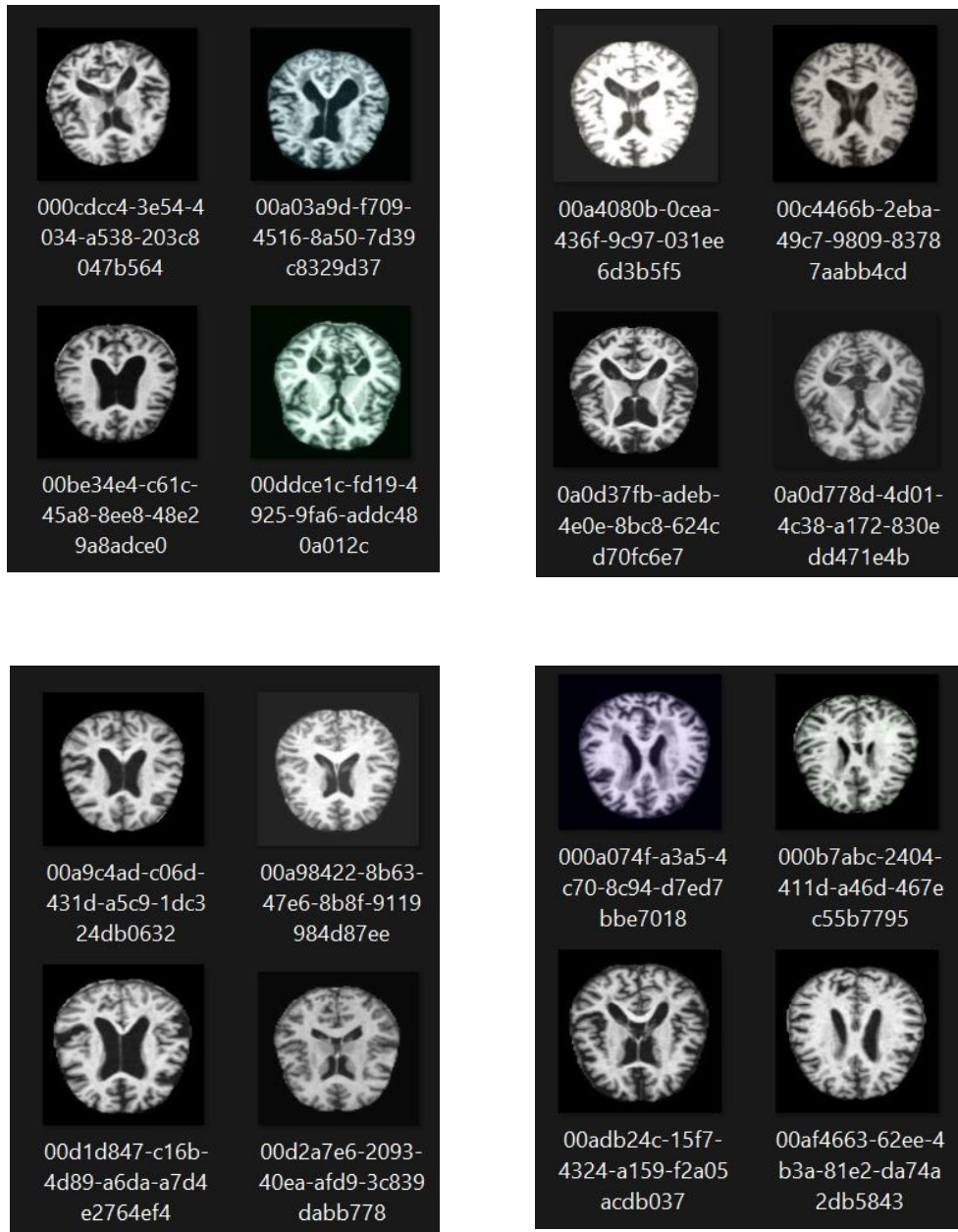


Figure 3.1 Training Files from Dataset for four different categories

Chapter – 4

SYSTEM ARCHITECTURE

4.1 ResNet18:

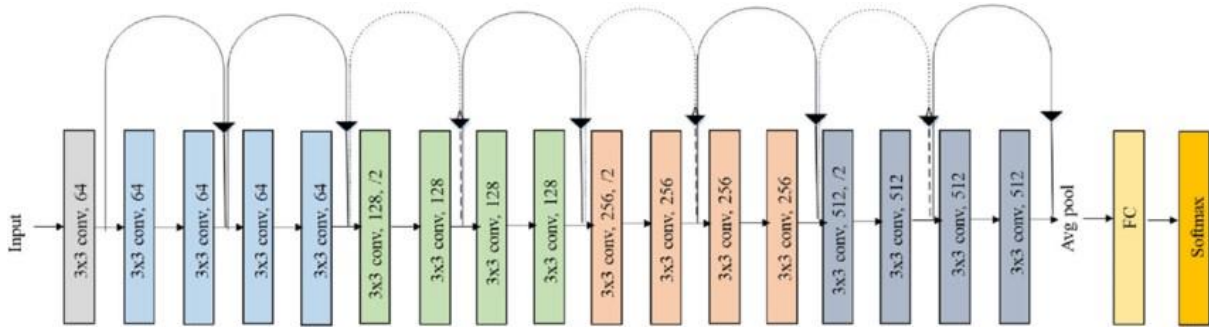


Figure 4.1 ResNet18

ResNet18 consists of 18 layers, including convolutional layers, activation layers, batch normalization using Max Pooling, and ReLU activation functions. It is relatively shallow compared to ResNet50 and is suitable for tasks where computational resources are limited.

ResNet18 is suitable for tasks where computational efficiency is a priority, such as real-time image classification on resource-constrained devices or systems with limited computational power. In the final layer, to identify the correct multivariate classes, the Softmax Activation Function is used.

Softmax is an activation function that scales numbers/logits into probabilities. The output of a Softmax is a vector (say v) with probabilities of each possible outcome. The probabilities in vector v sums to one for all possible outcomes or classes. The mathematical notation of the Softmax function is given in the next page.

$$S(y)_i = \frac{\exp(y_i)}{\sum_{j=1}^n \exp(y_j)}$$

where,

y	is an input vector to a softmax function, S. It consist of n elements for n classes (possible outcomes)
y_i	the i -th element of the input vector. It can take any value between -inf and +inf
$\exp(y_i)$	standard exponential function applied on y_i . The result is a small value (close to 0 but never 0) if $y_i < 0$ and a large value if y_i is large. eg • $\exp(55) = 7.69e+23$ (A very large value) • $\exp(-55) = 1.30e-24$ (A very small value close to 0) Note: $\exp(*)$ is just e^* where $e = 2.718$, the Euler's number.
$\sum_{j=1}^n \exp(y_j)$	A normalization term. It ensures that the values of output vector $S(y)_i$ sums to 1 for i -th class and each of them and each of them is in the range 0 and 1 which makes up a valid probability distribution.
n	Number of classes (possible outcomes)

Figure 4.2 SoftMax Function

Adam optimizer is used in improving the training phase. The advantage of Adam optimizer with some extensions to stochastic gradient descent is the Adaptive Gradient Algorithm (AdaGrad) which improves the performance on problems with small gradients and maintains a parameter learning rate. Another advantage of adam optimizer is Root Mean Square Propagation (RMSProp), which is based upon the averaging of the gradients for the learning rates of the weight parameter. The algorithm for the Adam Optimizer is given below this paragraph.

```

Require:  $\alpha$ : Stepsize
Require:  $\beta_1, \beta_2 \in [0, 1)$ : Exponential decay rates for the moment estimates
Require:  $f(\theta)$ : Stochastic objective function with parameters  $\theta$ 
Require:  $\theta_0$ : Initial parameter vector
 $m_0 \leftarrow 0$  (Initialize 1st moment vector)
 $v_0 \leftarrow 0$  (Initialize 2nd moment vector)
 $t \leftarrow 0$  (Initialize timestep)
while  $\theta_t$  not converged do
     $t \leftarrow t + 1$ 
     $g_t \leftarrow \nabla_{\theta} f_t(\theta_{t-1})$  (Get gradients w.r.t. stochastic objective at timestep  $t$ )
     $m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$  (Update biased first moment estimate)
     $v_t \leftarrow \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$  (Update biased second raw moment estimate)
     $\hat{m}_t \leftarrow m_t / (1 - \beta_1^t)$  (Compute bias-corrected first moment estimate)
     $\hat{v}_t \leftarrow v_t / (1 - \beta_2^t)$  (Compute bias-corrected second raw moment estimate)
     $\theta_t \leftarrow \theta_{t-1} - \alpha \cdot \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$  (Update parameters)
end while
return  $\theta_t$  (Resulting parameters)

```

Figure 4.3 Adam Optimizer

4.2 Custom ResNet64

The proposed system architecture is an extension of the ResNet (Residual Neural Network) architecture, which has proven to be highly effective in deep learning tasks, particularly in image classification. ResNet is known for its ability to train very deep neural networks effectively by utilizing residual blocks, which enable the learning of residual functions with reference to the layer inputs.

The extension presented here builds upon the ResNet architecture by incorporating additional layers and features to enhance its capabilities. The architecture begins with an initial convolutional layer followed by batch normalization and rectified linear unit (ReLU) activation, which serve to extract features from the input images. A max-pooling layer is then applied to reduce spatial dimensions while retaining important features.

The core of the extension lies in the series of residual blocks employed thereafter. Each residual block consists of two convolutional layers with batch normalization and ReLU activation applied after each convolution. The residual blocks allow the network to learn residual mappings, thereby facilitating the training of deeper networks while mitigating the vanishing gradient problem.

The extended architecture includes multiple stacks of residual blocks, with the number of filters increasing at deeper layers. This design choice allows the network to capture increasingly complex features as the information flows through the network. Additionally, certain residual blocks utilize strided convolutions to downsample the spatial dimensions, effectively increasing the receptive field of the network and reducing computational complexity.

To further regularize the network and prevent overfitting, dropout is applied before the final classification layer. Finally, a global average pooling layer is employed to aggregate spatial information from the previous layers, followed by a dense layer with softmax activation to produce class probabilities.

Overall, this extension of the ResNet architecture combines the strengths of residual learning with additional layers and regularization techniques, resulting in a powerful and efficient deep learning model suitable for image classification tasks.

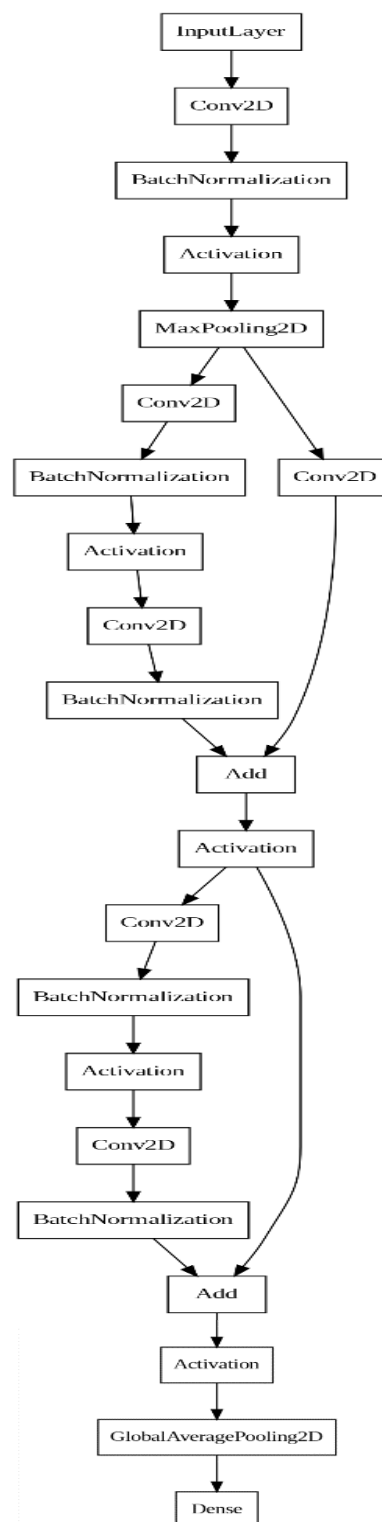


Figure 4.4 Custom ResNet64

4.3 CNN – SVM

The second proposed system architecture is a combination of a Convolutional Neural Network (CNN) and a Support Vector Machine (SVM) classifier, designed for image classification tasks. The architecture begins with data preprocessing to standardize input images before feeding them into the CNN model. This preprocessing step involves resizing the images to a standard size of (190, 200) pixels and converting them to float32 format to ensure consistency across the dataset. Standardizing the input images helps in preparing uniform data for subsequent processing.

Following preprocessing, the CNN model, serving as the feature extractor, is constructed. The CNN comprises several convolutional layers, each equipped with rectified linear unit (ReLU) activation functions to introduce non-linearity and facilitate learning of complex features. Max-pooling layers are inserted after convolutional layers to down sample feature maps and extract dominant features, while batch normalization layers normalize the input to each layer, thus stabilizing and accelerating the training process. The CNN architecture culminates with a GlobalAveragePooling2D layer, which aggregates spatial information and reduces the feature maps to a single vector, making them compatible with the SVM classifier. Additionally, a dropout layer is incorporated to prevent overfitting by randomly deactivating a fraction of neurons during training.

In the subsequent stage, the standardized features extracted by the CNN model are fed into the SVM classifier. These features are further standardized using a StandardScaler to ensure zero mean and unit variance, enhancing the SVM's performance. The SVM classifier, initialized with a radial basis function (RBF) kernel, is trained to classify feature vectors into different classes. During training,

the SVM learns to separate feature vectors belonging to distinct classes with a maximum margin, thus improving classification accuracy.

During the testing phase, the trained CNN model is employed to extract features from validation images, which are then scaled using the same StandardScaler used for training data. The scaled features are inputted into the trained SVM classifier to predict labels for validation images. Finally, the accuracy of the SVM classifier is calculated by comparing predicted labels with true labels from the validation dataset, providing a measure of the model's performance. This architecture offers a robust and efficient solution by combining the feature extraction capabilities of CNNs with the classification prowess of SVMs, resulting in accurate and reliable image classification.

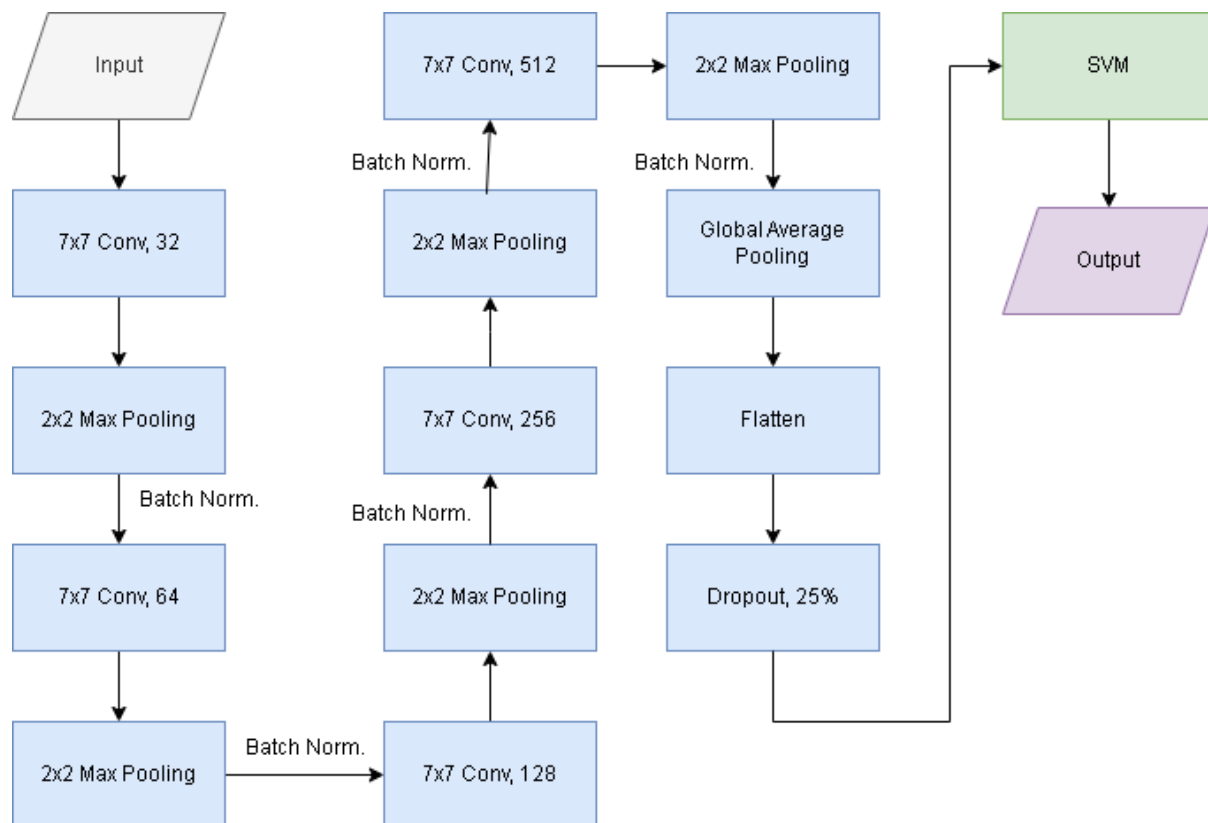


Figure 4.5 CNN-SVM Architecture

Chapter 5

SYSTEM REQUIREMENTS

5.1 Introduction

Making the model predict the class of Alzheimer's is greatly dependent on the set of hardware/software tools used in designing and training the model. As this project is highly sensitive and requires careful attention to details, the choice of tools used in development is made after considering various alternatives and studying the advantages and disadvantages of each.

5.2 Hardware Requirements

5.2.1. Computational Device:

The system requires a computational device capable of running complex algorithms for image processing and artificial intelligence tasks.

5.2.2. Graphics Processing Unit (GPU):

A GPU with sufficient computational power is essential for accelerating the training and inference processes of the AI model.

5.2.3. Storage:

Adequate storage capacity is necessary to store datasets, trained models, and intermediate results.

5.2.4. Network Infrastructure:

A stable network infrastructure is crucial for data transfer between the computational device and any remote servers or cloud-based services used in the system.

5.3 Software Requirements

5.3.1. Machine Learning Framework:

The system relies on a machine learning framework, TensorFlow for developing and training the custom ResNet model. TensorFlow offers a comprehensive ecosystem for building and deploying machine learning models, including complex CNNs, with its extensive collection of tools and libraries.

5.3.2. Image Processing Libraries:

Keras has inbuilt functions to modify the input image to suit the model's requirements. Additionally, OpenCV library was used to test the functioning of the model, which was used to load the image and perform any necessary rescaling.

5.3.3. Development Environment:

Google Colaboratory, a hosted Jupyter notebook is the prime development environment for model construction, training and evaluation. Using the T4 GPU bundled as a part of the plan, the model construction and training was handled efficiently.

5.3.4. User Interface:

HTML, CSS are the primary front-end components used in developing the user interface. Additionally, React JS is utilized to improve the readability and adopt to the “Don’t repeat yourself” coding structure.

Chapter 6

SYSTEM DESIGN AND IMPLEMENTATION

6.1 OVERALL DESIGN

6.1.1 ResNet64 Model:

The ResNet (Residual Network) model is designed to classify brain MRI images into four categories: Mild Demented, Moderate Demented, Non Demented, and Very Mild Demented. The architecture of the ResNet model is inspired by the ResNet18 architecture, modified to suit the specific requirements of the Alzheimer's disease classification task.

6.1.2 ResNet64 Model Architecture:

The ResNet model consists of convolutional layers with residual blocks and global average pooling. It starts with a convolutional layer followed by batch normalization and ReLU activation. The model includes residual blocks containing two convolutional layers with skip connections to mitigate vanishing gradient problems. After the convolutional layers, there is a global average pooling layer followed by a dense layer with softmax activation to produce class probabilities.

6.1.3 Training and Evaluation:

The ResNet model is trained using the augmented Alzheimer's dataset. Images are preprocessed by resizing to 200x190 pixels and normalization. During training, the model learns to classify images into appropriate categories using the training dataset. The performance of the model is evaluated using the validation dataset to measure accuracy and loss.

6.1.4 CNN-SVM Model:

The Support Vector Machine (SVM) model is employed as a classifier to predict the class labels of brain MRI images based on features extracted from the CNN model instead of the dense neuron layer.

6.1.5 CNN-SVM Model Architecture:

Features are extracted from the CNN model using the penultimate layer's output. These features are then standardized using StandardScaler to ensure consistent scaling across features. An SVM classifier with a radial basis function (RBF) kernel is trained using the extracted and standardized features.

6.1.6. Training and Evaluation:

The SVM model is trained using the extracted features from the training dataset. StandardScaler is fit on the training features and applied to both training and validation features. The SVM classifier is trained on the standardized features and evaluated using the validation dataset to assess its accuracy.

6.2 RESNET64 MODEL IMPLEMENTATION

6.2.1 Model Architecture:

The ResNet model is implemented using TensorFlow and Keras. It comprises convolutional layers, residual blocks, batch normalization, and a global average pooling layer, accumulating a total 64 layers. The model is designed to handle input images of size 200x190 pixels and output class probabilities for the four categories.

6.2.2 Training and Testing:

The ResNet64 model is trained using the augmented Alzheimer's dataset. Images are preprocessed, resized, and normalized before being fed into the model. During training, the model learns to classify images into the appropriate categories. The model's performance is evaluated using the validation dataset to measure accuracy and loss.

6.3 CNN-SVM MODEL IMPLEMENTATION

6.3.1. Feature Extraction:

Features are extracted from the ResNet model using the penultimate layer's output. These features represent the learned representations of the input images. Feature extraction is performed on both training and validation datasets.

6.3.2 Standardization:

Extracted features are standardized using StandardScaler. StandardScaler is fit on the training features and applied to both training and validation features to ensure consistent scaling.

6.3.3 Training and Testing:

The standardized features are used to train an SVM classifier with an RBF kernel. The SVM classifier is trained on the training features and labels. The trained SVM classifier is evaluated on the validation dataset to assess its accuracy in predicting class labels.

6.4 CONCLUSION

The ResNet64 architecture proves to learn better features and has better validation accuracy as compared to the CNN-SVM hybrid model. The ResNet's convolutional layers and residual structure greatly improves the learning phase, while the CNN-SVM hybrid is not accurate, as the features extracted by the CNN are not well suited for SVM as it is suited for vector based discrimination.

Chapter 7

RESULT AND ANALYSIS

7.1 MRI-Based Alzheimer's Classification

7.1.1 ResNet64 Model Performance Metrics

Our convolutional neural network (CNN) model, based on the ResNet64 architecture, was trained to classify Alzheimer's disease into four categories: mild, moderate, very mild and No dementia based on MRI scans of the brain. The following are the key performance metrics of our model:

- **Accuracy:** The overall accuracy of our model on the test dataset is 94.73%.
- **Precision:** The weighted average of precision for all classes on the test dataset is 0.94.
- **Recall:** The weighted average of recall for all classes on the test dataset is 0.94.
- **F1:** The overall weighted average F1 scores of our model is 0.94.

The model was trained for multiple epochs. Considering the size of the available dataset, the epochs were limited to at most 30 to prevent overfitting the model. The accuracy curve for the training and validation of the model for different epochs can be referred in **Figure 7.1** and **Figure 7.2**.

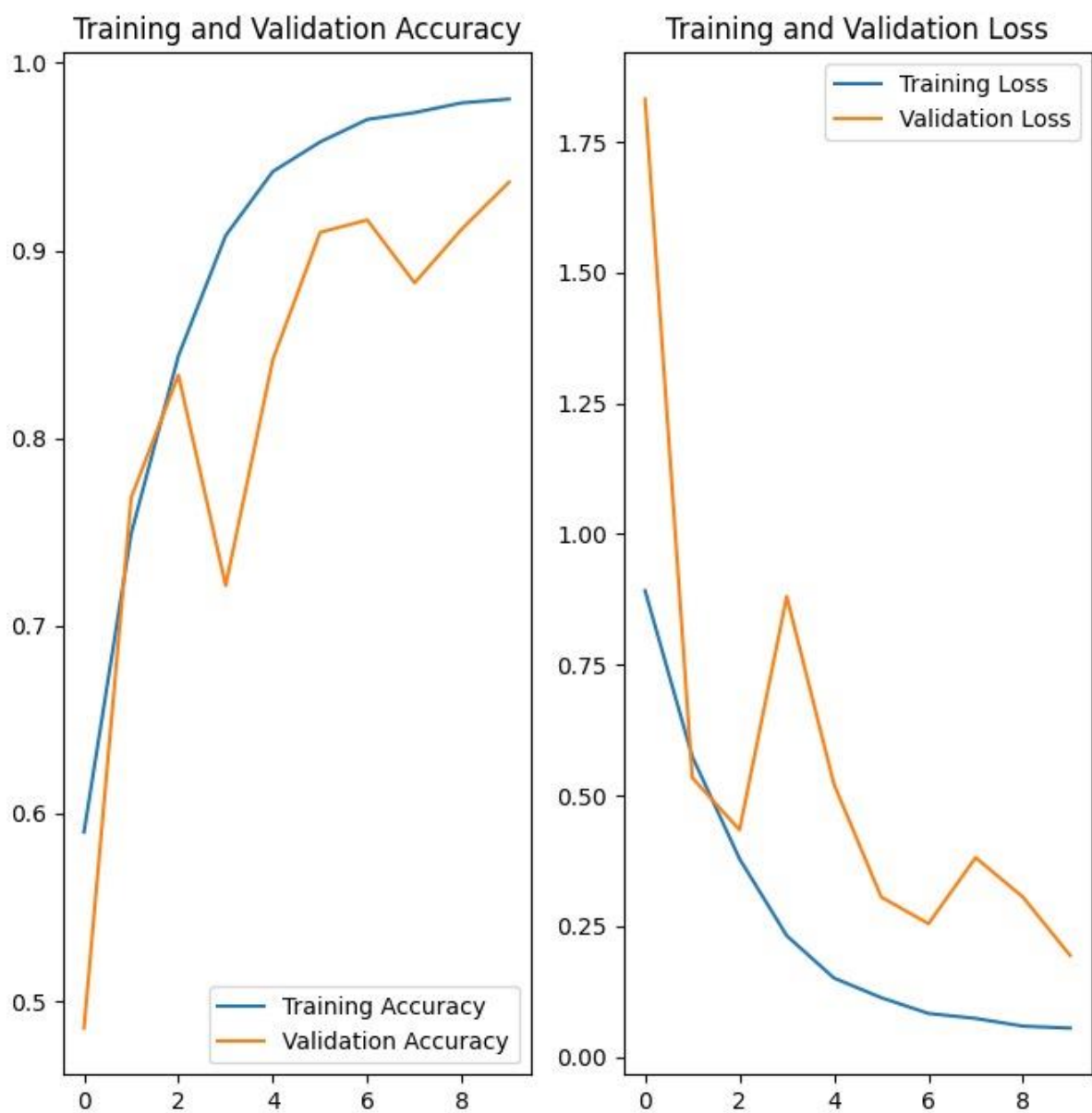


Figure 7.1 – Accuracy Curve after 10 Epochs

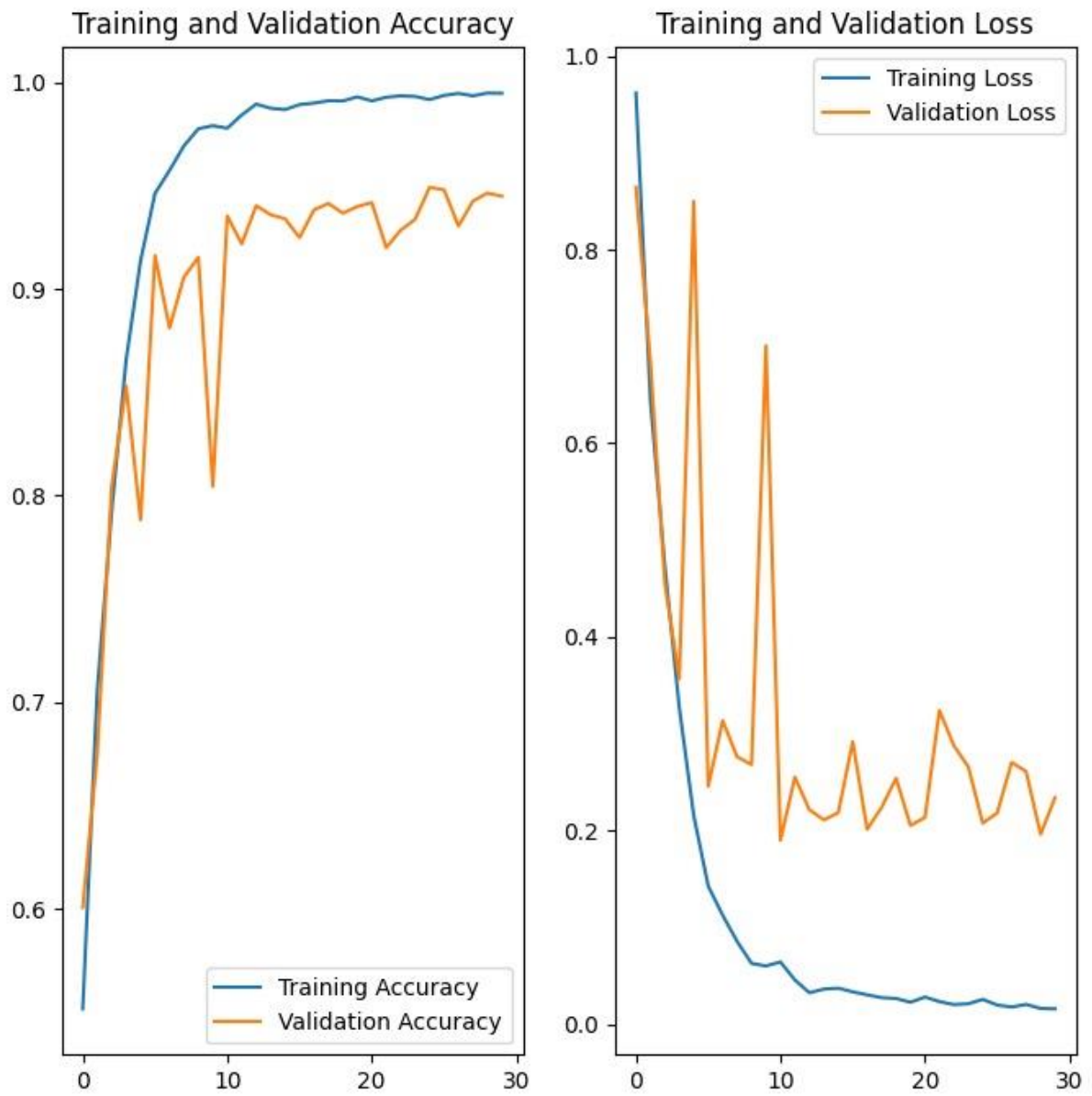


Figure 7.2 – Accuracy Curve after 30 Epochs

7.2 Hybrid CNN-SVM Model for Alzheimer's Disease Classification

Upon implementing our hybrid convolutional neural network (CNN) model integrated with Support Vector Machine (SVM) for Alzheimer's disease classification, we achieved an accuracy of 70%. This amalgamation aimed to synergize the feature extraction capabilities of CNNs from MRI scans with the classification prowess of SVMs. While the attained accuracy denotes a commendable performance, it also underscores avenues for further refinement and optimization. Future endeavors could explore parameter fine-tuning, incorporation of additional features or datasets, and utilization of advanced methodologies to augment classification accuracy. Notwithstanding the moderate accuracy, this hybrid model signifies a progressive step in leveraging machine learning methodologies for Alzheimer's disease diagnosis, thereby laying a robust foundation for ongoing research and development in this pivotal domain of healthcare.

7.3 Performance Evaluation

7.3.1 CNN-SVM Model for Alzheimer's Disease Classification

The CNN-SVM model was evaluated on a validation dataset, achieving a validation accuracy of 70.14%. This performance is further analyzed using precision, recall, and F1-score metrics, along with a confusion matrix, to provide a comprehensive understanding of the model's classification abilities.

Validation Accuracy:

- The model achieved an overall validation accuracy of 70.14%. This metric indicates the proportion of correctly classified instances out of the total instances in the validation set.

Classification Report:

- The classification report provides detailed performance metrics for each class, including precision, recall, and F1-score.

	precision	recall	f1-score	support
Mild Demented	0.63	0.79	0.70	1801
Moderate Demented	0.89	0.91	0.90	1319
Not Demented	0.73	0.66	0.69	1942
Very Mild Demented	0.61	0.50	0.55	1764

Accuracy:

- Overall accuracy across all classes is 70.14%.

Macro Average:

- **Precision:** 0.71
- **Recall:** 0.71
- **F1-Score:** 0.71

Weighted Average:

- **Precision:** 0.70
- **Recall:** 0.70
- **F1-Score:** 0.70

Confusion Matrix:

- The confusion matrix provides insight into the specific misclassifications made by the model.

Predicted	Class 0	Class 1	Class 2	Class 3
Actual 0	1417	69	90	225
Actual 1	86	1203	7	23
Actual 2	314	32	1278	318
Actual 3	450	46	369	869

Classes Analysis:

- **Class 0 (Non-Demented):**

- The model demonstrates good recall (0.79), meaning it correctly identifies 79% of actual non-demented cases. However, the precision is slightly lower (0.63), indicating some misclassification of other classes as non-demented.

- **Class 1 (Very Mild Demented):**

- This class shows the highest precision (0.89) and recall (0.91), suggesting the model is very effective at identifying very mild dementia and rarely misclassifies it.

- **Class 2 (Mild Demented):**

- The recall for mild dementia is moderate (0.66), with a precision of 0.73, indicating the model has a reasonable balance in identifying mild dementia cases but still has room for improvement.

- **Class 3 (Moderate Demented):**

The model struggles more with moderate dementia, as indicated by the lower recall (0.50) and precision (0.61). This suggests a higher rate of misclassifications between moderate dementia and other classes.

Overall, the macro and weighted averages of precision, recall, and F1-score are around 0.70, indicating a consistent performance across different metrics.

The confusion matrix reveals that misclassifications predominantly occur between classes with similar stages of dementia, highlighting the challenge in distinguishing between varying severity levels of Alzheimer's disease.

7.3.2 Custom ResNet Model for Alzheimer's Disease Classification

The performance of the custom ResNet CNN model was evaluated using precision, recall, F1-score, accuracy, and a confusion matrix. The following table summarizes the performance metrics for each class:

Precision, Recall, F1-Score, and Support

	precision	recall	f1-score	support
Mild Demented	0.97	0.95	0.96	1801
Moderate Demented	1.00	0.99	1.00	1319
Not Demented	0.97	0.87	0.92	1942
Very Mild Demented	0.84	0.96	0.90	1764

Overall Metrics

- Accuracy: 94%
- Macro Average:
 - Precision: 0.94
 - Recall: 0.94
 - F1-Score: 0.94
- Weighted Average:
 - Precision: 0.94
 - Recall: 0.94
 - F1-Score: 0.94

Confusion Matrix

The confusion matrix provides a detailed breakdown of the model's predictions versus actual labels:

	0	1	2	3
0	1705	1	13	82
1	5	1308	0	6
2	26	0	1686	230
3	29	1	39	1665

0 - Mild Demented

1 - Moderate Demented

2 - Not Demented

3 - Very Mild Demented

Interpretation of Metrics

Class 0:

- Precision: 0.97, indicating that 97% of the instances predicted as class 0 are correct.
- Recall: 0.95, meaning 95% of the actual class 0 instances are correctly identified.
- F1-Score: 0.96, providing a balance between precision and recall.

Class 1:

- Precision: 1.00, indicating perfect prediction accuracy for instances of this class.
- Recall: 0.99, meaning 99% of actual class 1 instances are correctly identified.
- F1-Score: 1.00, showing excellent model performance for this class.

Class 2:

- Precision: 0.97, suggesting high accuracy in predicting this class.
- Recall: 0.87, indicating some misclassifications.
- F1-Score: 0.92, reflecting a good balance between precision and recall.

Class 3:

- Precision: 0.84, meaning 84% of the instances predicted as class 3 are correct.
- Recall: 0.96, showing that the majority of actual class 3 instances are correctly identified.
- F1-Score: 0.90, indicating good overall performance for this class.

Misclassifications:

The confusion matrix shows 82 instances that were actually Mild Demented (Class 0) were misclassified as Very Mild Demented (Class 3). There are also 230 instances that were truly Not Demented (Class 2) but were incorrectly identified as Very Mild Demented (Class 3). This is a more serious misclassification, as it suggests the model is mistaking healthy individuals for those with some level of dementia.

How the MoCA Test Can Help:

The MoCA test is a brief, standardized assessment used by doctors to evaluate cognitive function. By incorporating the MoCA test alongside the model's predictions, doctors can gain additional insights and potentially avoid these misdiagnoses:

- For instances where the model predicts Very Mild Demented (Class 3), a MoCA test can provide a more definitive diagnosis, especially for those with a history of dementia in the family or showing other cognitive concerns.
- The MoCA test can be particularly helpful when the model's prediction is Very Mild Demented (Class 3) for patients classified as Not Demented (Class 0) by the model. The MoCA test can offer a second layer of verification before making a diagnosis.

Overall, while the model shows promising performance, using the MoCA test in conjunction with the model's predictions can improve diagnostic accuracy and ensure better patient care, especially when dealing with borderline or uncertain classifications.

7.4 Interpretation of Results

Our model exhibits promising performance in effectively classifying Alzheimer's disease severity based on MRI scans. The high accuracy achieved underscores its proficiency in discerning between various stages of Alzheimer's with notable confidence. Furthermore, contrasted with the lower accuracy observed in the SVM model, the robustness of the ResNet architecture emerges as evidently more reliable in this context.

7.5 Moca Test Integration

7.5.1 Feature Importance

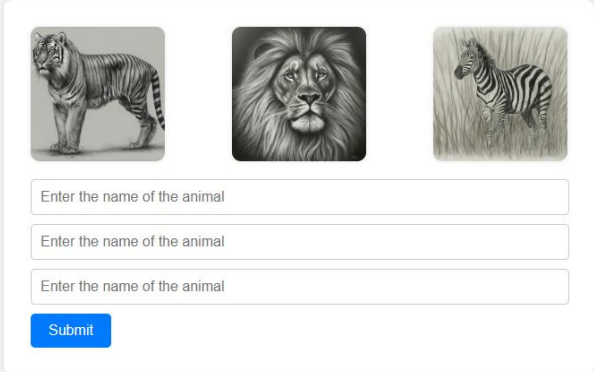
In addition to MRI scans, our project incorporates the Montreal Cognitive Assessment (MoCA) test results as an additional feature to enhance the Alzheimer's disease detection process. Through feature importance analysis, we found that the MoCA test scores significantly contribute to the detection of Alzheimer's disease in the elderly.

Below are the example MoCA test questions and the web interface that administers the test and allows the healthcare professional to use the model for processing the MRI of the patient to get an accurate prediction.

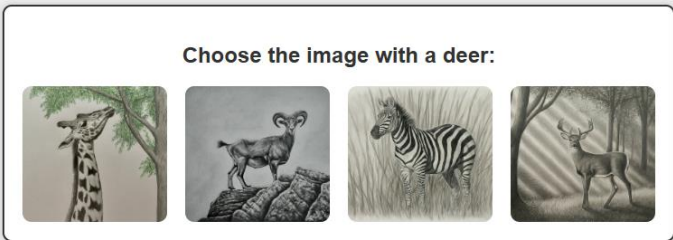
The questions include assessment of orientation to time, orientation to place, registration, attention and calculation, recall, language, repetition and complex commands.

7.5.2 MoCA Test Questions

- **Naming:** The person will be shown pictures of three animals and asked what type of animal they are. Then they will be asked to associate the name of an animal to the picture of the animal from pictures of four animals.



The interface for the Naming test consists of a white box on a light gray background. At the top, there are three square images: a tiger, a lion, and a zebra. Below the images are three text input fields, each with the placeholder text "Enter the name of the animal". At the bottom of the box is a blue button labeled "Submit".

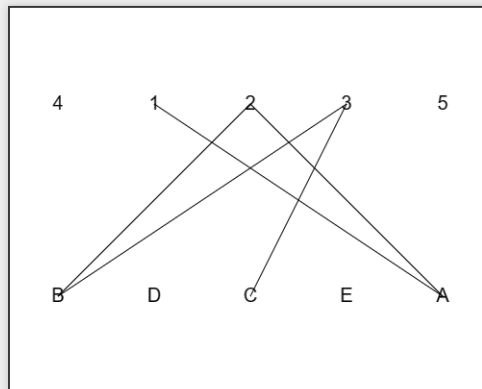


The interface for the Naming test consists of a white box on a light gray background. At the top, there is a text prompt: "Choose the image with a deer:". Below the prompt are four square images: a giraffe, a goat, a zebra, and a deer. The box is bordered by a thin black line.

- **Executive and visuospatial function:** You will first be given a picture with numbered dots (1, 2, 3, 4, 5) and lettered dots (A, B, C, D, E) and asked to connect them sequentially, alternating numbers and letters. Next, you will be asked to draw a dial clock that reads 30 minutes past 06:00.

Please draw a line going from a number to a letter in ascending order

Begin here [point to (1)] and draw a line from 1 then to A then to 2 and so on. End here [point to (E)].



Set the clock to time 6:30



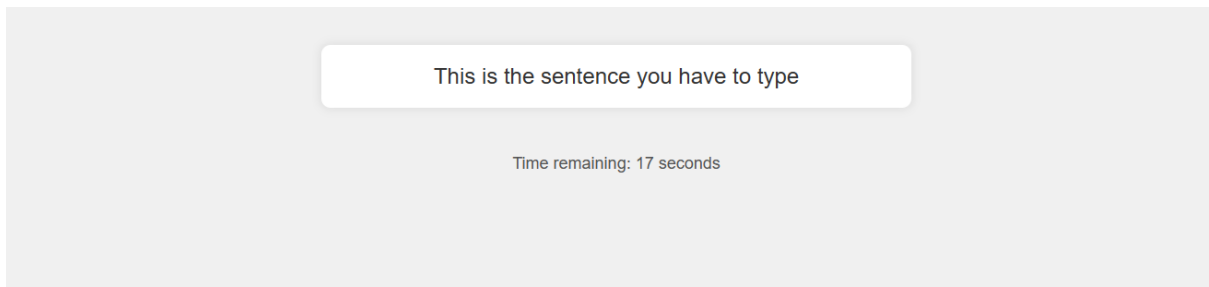
Submit

- **Orientation:** The patient will be asked about the date, month, year, day, city, or place they are in.

What is the year, season, date, day, month, and year? (YYYY, Season, MM/DD/YYYY, Day, Month, YYYY)

Next

- **Delayed Recall:** The patient will be given a sentence to see and asked to repeat it back back within a minute.

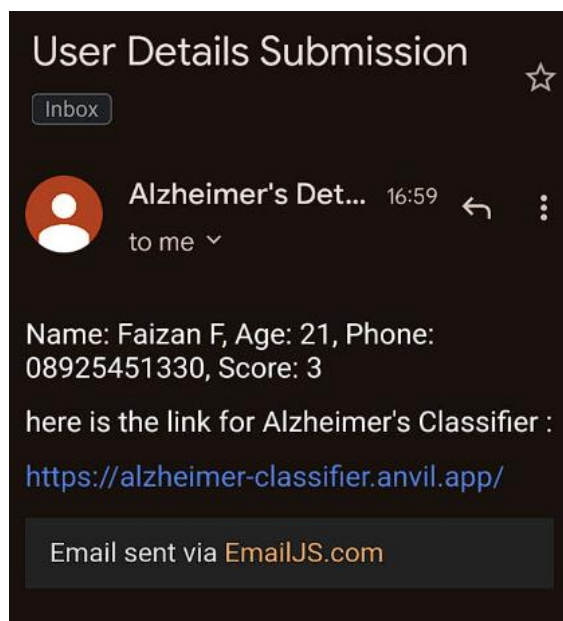


- **Score Assessment** – Finally the scores tallied from the questions will be checked with a threshold value. If the patient scores a low score for their age group, then an email with the patient's details and score will be sent to the medical professional who administers the test.

Your final score is:

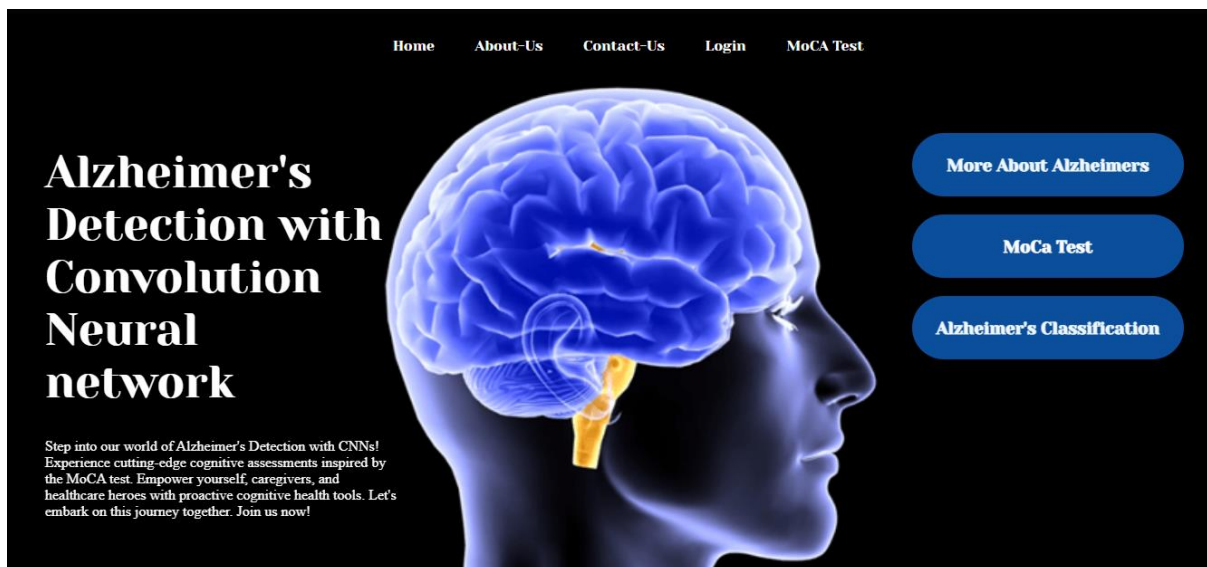
2/9

Play Again

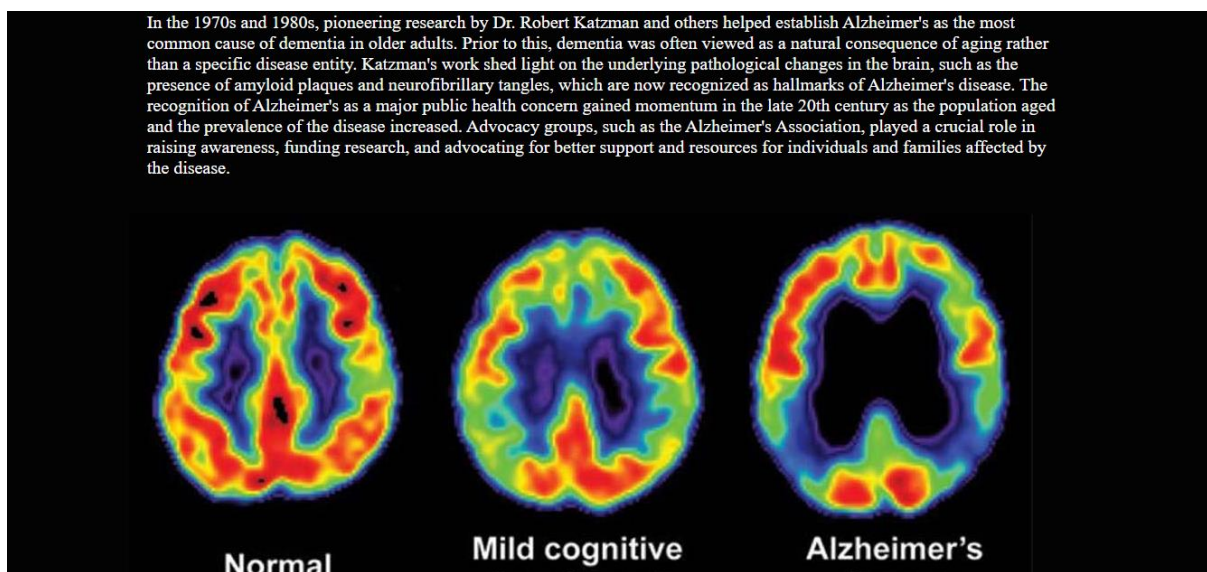


7.5.3 Webpage Integration of MoCA test and Classification Model

7.5.3.1 User-Interface



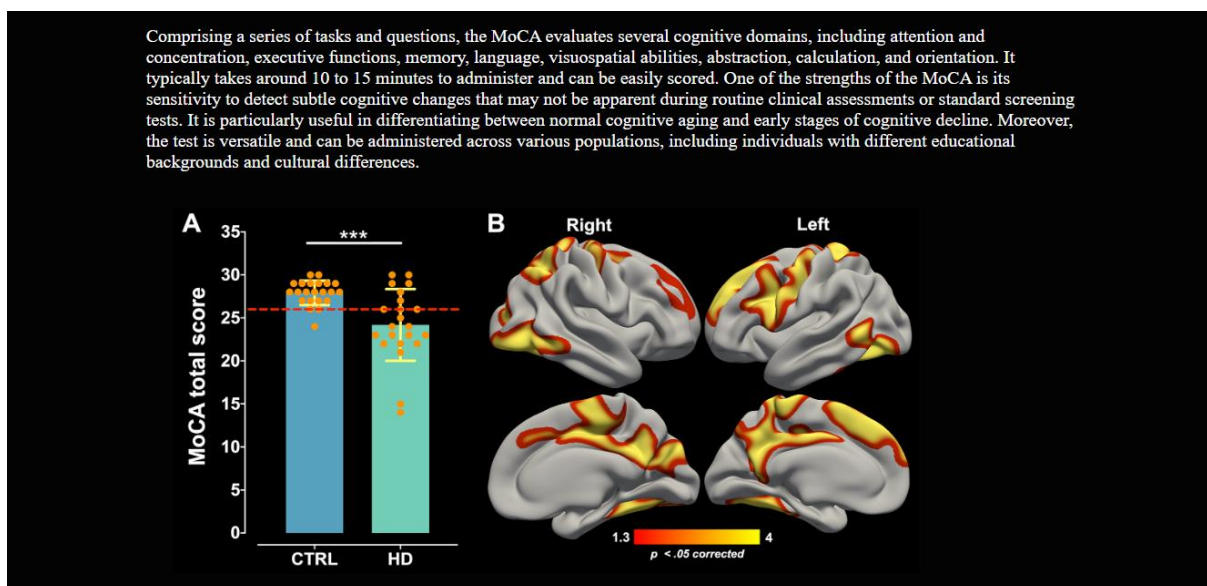
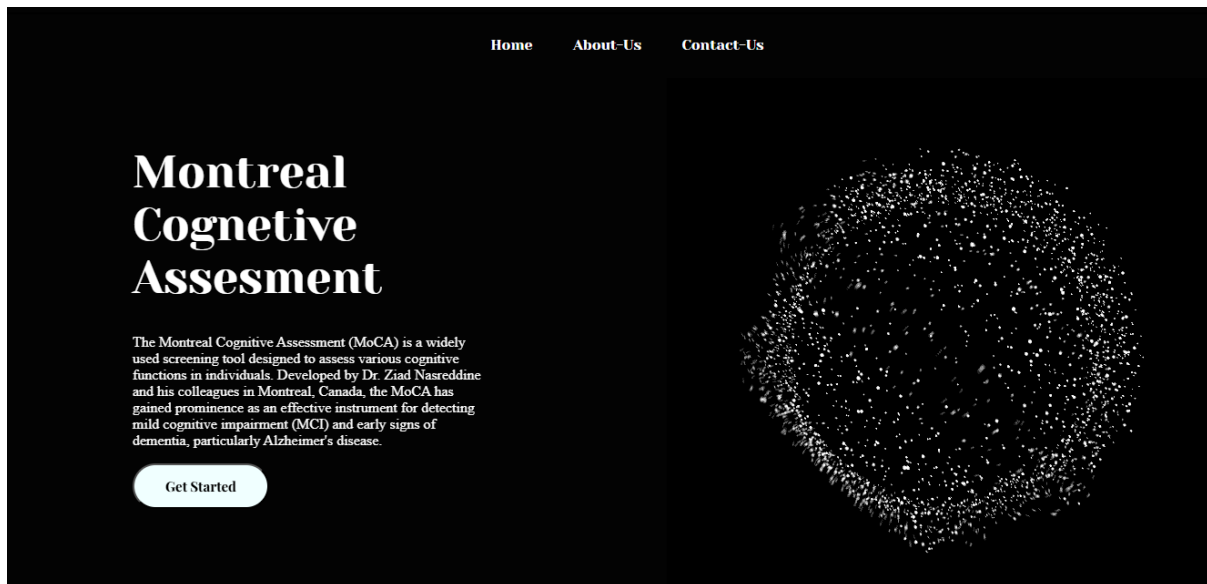
Landing Page: A landing page is created to provide users with various options, such as taking the MOCA test, accessing Alzheimer's classification tools, and learning more about Alzheimer's disease.



A section offering basic knowledge about Alzheimer's disease is also provided to educate users and increase awareness.

MOCA Test Page:

- A dedicated page for the MOCA test is created where users can take the test. This page also provides basic knowledge about the MOCA test, including its purpose, structure, and importance in cognitive assessment.



Centre for AD

Upload an MRI to classify the patient into one of the three stages of Alzheimer or healthy if not.

 Upload

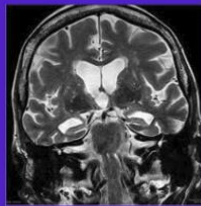
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Post-Test Navigation:

- Based on the results of the MOCA test, users are directed to a specialized model that classifies Alzheimer's disease. This model provides an analysis and detailed classification of the user's cognitive health status.

Centre for AD

Upload an MRI to classify the patient into one of the three stages of Alzheimer or healthy if not.



Moderate Demented

 1 file selected

7.6 Combined Model Performance

By integrating MoCA test results, our model achieves enhanced accuracy and robustness in Alzheimer's disease prediction and classification. The model combined with the test scores provide a comprehensive approach by leveraging

both structural (MRI) and functional (MoCA test) information, resulting in more reliable predictions.

7.7 Clinical Implications

The successful development and validation of our CNN model hold significant clinical implications. Accurate classification of Alzheimer's disease severity at an early stage can facilitate timely intervention and personalized treatment strategies. Moreover, the integration of MoCA test results enhances the diagnostic process, enabling professionals to make informed decisions regarding patient care.

Our CNN model, integrated with the MoCA test, offers a valuable tool for Alzheimer's disease diagnosis and prognosis. Further refinement and validation of the model hold promise for advancing early detection and management strategies in clinical practice.

Chapter 8

CONCLUSION AND FUTURE WORK

8.1 Conclusion

Our study demonstrates the potential of advanced machine learning techniques, particularly convolutional neural networks (CNNs), in accurately detecting and classifying Alzheimer's disease severity based on MRI scans. The high accuracy achieved underscores the effectiveness of our CNN-based model in distinguishing between different stages of Alzheimer's disease. Notably, compared to alternative methods such as Support Vector Machines (SVMs), our CNN model has shown superior performance, emphasizing its reliability in medical applications.

8.2 Future Work

Looking over the current boundaries of this study, the next step is envisioning a comprehensive platform poised to revolutionize medical diagnostics and patient care. In pursuit of this vision, we outline the following avenues for future implementation and enhancement:

1. **Expansion of Disease Coverage:** Our ambition is to extend the capabilities of our platform to encompass a broader spectrum of diseases beyond Alzheimer's. By leveraging CNN-based algorithms, we aim to facilitate accurate diagnosis and prognosis for various medical conditions, thus improving healthcare accessibility and outcomes.
2. **Evaluating Model Performance and Optimize the Model:** Our current model possesses the ability to accurately classify the MRI images into three stages of Alzheimer's and one Non-demented stage with a high accuracy and precision scores. The ResNet model's overall performance may be improved with further statistical analysis and optimization techniques.

3. **Data Warehousing and Management:** Implementing a robust data warehousing infrastructure will be pivotal in securely storing and managing diverse medical datasets, including MRI scans and associated patient information. This will enable seamless access, analysis, and integration of data, enhancing the efficiency and reliability of our platform.
4. **Automated Model Training and Updates:** Embracing automated model training pipelines will ensure that our algorithms remain up-to-date with the latest advancements in medical imaging and machine learning. By continuously refining and optimizing our models, we can enhance their performance and adaptability to evolving clinical needs.
5. **Efficiency and Performance Optimization:** Given the critical nature of medical applications, optimizing the efficiency and performance of our platform is paramount. Through rigorous testing, optimization, and validation processes, we aim to enhance the speed, accuracy, and reliability of our algorithms, thereby maximizing their clinical utility.
6. **Online Doctor Support and Collaborative Care:** Introducing features for online doctor support and collaborative care will empower healthcare professionals to provide timely and personalized interventions to patients. By facilitating seamless communication and information exchange between patients and caregivers, we can foster a patient-centric approach to healthcare delivery.

By embracing these initiatives, we aspire to not only consolidate our position at the forefront of medical AI but also to catalyse transformative advancements in healthcare technology. Through innovation, collaboration, and a steadfast commitment to improving patient outcomes, we endeavour to realize our vision of a future where cutting-edge technology empowers individuals to lead healthier, more fulfilling lives.

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